Forces 2

Ch 6 HRW

Friction and circular motion

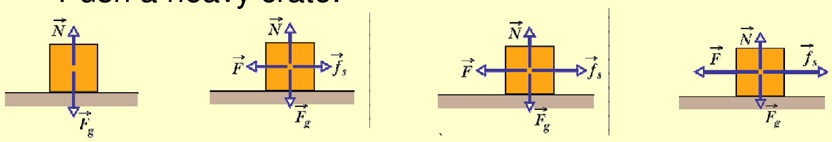
Friction

- Static Friction
- Kinetic Friction
- Drag
- Overcoming friction accounts for much of the energy consumed every day.
- Without friction nothing much would happen either.
  - Walk, ride bicycle, write with a pen/pencil.

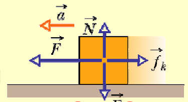
2

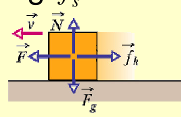
Friction

- Book Slides across a long table
  - Book slows then stops.
  - Acceleration parallel to table surface, opposite to v.
- Book being pushed along table with constant v.
  - Is the force you apply the only horizontal force acting on the book?
- Push a heavy crate.



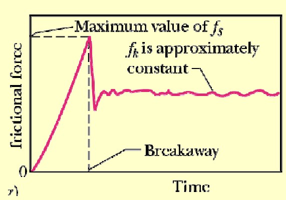
- Increasing force $\vec{F}$ and increasing $\vec{f}_s$- STATIC FRICTION



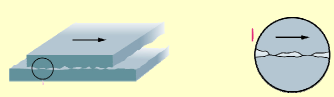
- Acceleration

Constant v.
- $\vec{f}_k$ KINETIC FRICTION

Friction

- Kinetic friction $\vec{f}_k < \vec{f}_s$.
- For constant v – applied force needs to be reduced when motion starts.



- Read Section 6-2 for more info on what is happening on a microscopic level.



Properties of Friction

- Dry, unlubricated object presses against a dry, unlubr-icated surface and a force $\vec{F}$ attempts to slide the body along the surface, the resulting frictional force has 3 properties:-
- Property 1:
  – If the body does not move, then the static frictional force and the component of the applied force that is parallel to the surface balance each other. $\vec{F} = \vec{f}_s$
- Property 2:
  – $\left|\vec{f}_{s,\max}\right| = \mu_s N$, where μ_s is coefficient of static friction, N is normal force. If $\vec{F} > \vec{f}_{s,\max}$, then body begins to slide.
- Property 3:
  – When body starts slide, $\left|\vec{f}_k\right| = \mu_k N$, where μ_k is coefficient of kinetic friction, N is normal force.

Properties of Friction

- μ_s and μ_k are dimensionless
- Determined experimentally
- Depend on the properties of the body and the surface
  – μ_s between egg on teflon coated pan is 0.04
  – μ_s between rock-climbing shoes and rock is about 1.2
- Assume that μ_k is independent of the speed at which the sliding occurs.

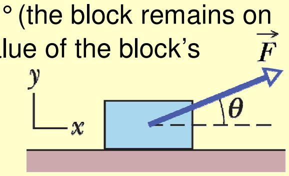
Checkpoint 1

- A block lies on a floor.
  - (a) What is the magnitude of the frictional force on it from the floor?
  - (b) If a horizontal force of 5 N is now applied to the block, but the block does not move, what is the magnitude of the frictional force on it?
  - (c) If the maximum value of the static frictional force on the block is 10 N, will the block move if the magnitude of the horizontally applied force is 8 N?
  - (d) If it is 12 N?
  - (e) What is the magnitude of the frictional force in part (c)?

7

Sample Problem p120

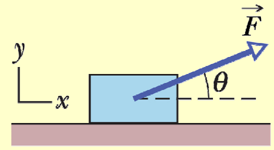
- In the Figure, a block of mass m = 3.0 kg slides along a floor while a force $\vec{F}$ of magnitude 12.0 N is applied to it at an upward angle θ. The coefficient of kinetic friction between the block and the floor is μ_k = 0.40 . We can vary θ from 0 to 90° (the block remains on the floor). What θ gives the maximum value of the block's acceleration magnitude a?

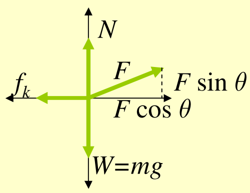


8

Sample Problem p120

- What θ gives the maximum value of the block's acceleration magnitude a?



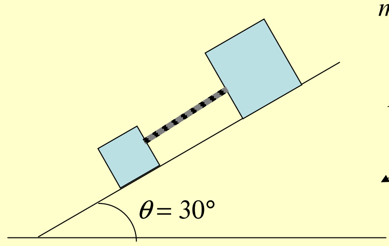


$$a = \frac{F\cos\theta - \mu_k N}{m}$$
$$= \frac{F\cos\theta - \mu_k\left(mg - F\sin\theta\right)}{m}$$

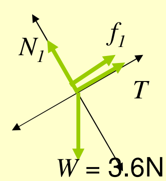
9

Problem

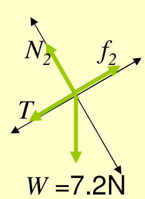
- Two blocks, of weights 3.6 N and 7.2 N are connected by a massless string and slide down a 30° inclined plane. The coefficient of kinetic friction between the lighter block and the plane is 0.10; that between the heavier block and the plane is 0.20. Assuming that the lighter block leads, find (a) the magnitude of the acceleration of the blocks and (b) the tension in the string.



$m_1 = 0.36$ kg



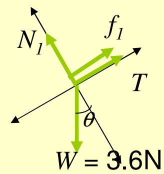
$m_2 = 0.74$ kg

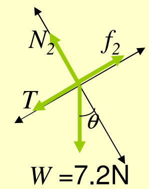


10

Problem

- Assuming that the lighter block leads, find (a) the magnitude of the acceleration of the blocks and (b) the tension in the string.

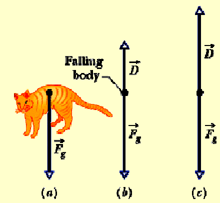




Drag force and Terminal Speed

- When there is a relative velocity between a fluid and a body, the body experiences a drag force, $\vec{D}$.
- Fluid – any thing that can flow – liquid or gas.
- This chapter –
  - Fluid is air and body is blunt (e.g. ball) compared to a sharp object (e.g. javelin)
  - Motion is fast and air becomes turbulent behind body.
- $D = \frac{1}{2} C \rho A v^2$
  - C – drag coefficient, ρ – density of air, A – effective cross-sectional area (perpendicular to velocity v.)
- $\vec{D}$ is directed opposite to direction of motion

Drag force and Terminal Speed



- $D = \frac{1}{2} C \rho A v^2$
- Blunt body falling from rest through air
- $\vec{D}$ is upward
- Magnitude gradually increases from zero as v increases.
- Forces acting on body (Newton 2):

$$D - W = ma$$

- If body falls long enough, D will increase until $D = W$.
- Then $a = 0$ – terminal speed, v_t – a constant.

$$\tfrac{1}{2} C \rho A v^2 - mg = ma = 0$$

$$\tfrac{1}{2} C \rho A v^2 = mg \qquad \Rightarrow v = \sqrt{\frac{2mg}{C\rho A}}$$

13

Sample Problem p123

- A raindrop with radius R = 1.5 mm falls from a cloud that is at a height h = 1200 m above ground. The drag coefficient C for the drop is 0.60 . Assume the drop is spherical throughout its fall. The density of water ρ_w is 1000 kg/m^3, and the density of air ρ_a is 1.2 kg/m^3.
  - What is the terminal speed of the drop?
  - What would be the drop's speed just before impact if there were no drag force?

14

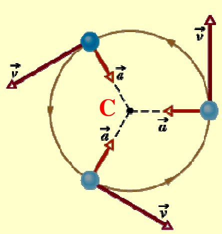
Problem 39

- Calculate the ratio of the drag force on a jet flying at 1000 km/h at an altitude of 10 km to the drag force on a prop-driven transport flying at half the speed and altitude of 5.0 km Assume that the planes have the same effective cross-sectional area and drag coefficient C. The density of air at an altitude of 10 km is 0.38 kg/m^3 and at 5 km it is 0.67 kg/m^3.

Uniform Circular Motion

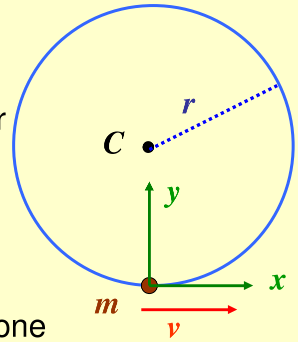
- Body moving in a circle with radius r at constant speed v
- $a = \frac{v^2}{r}$
- Net force directed toward C.

$$F_{net} = m\frac{v^2}{r}$$



- What causes the circular motion? This is NOT a new force!
- Examples:
  - Rounding a curve in a car.
  - Orbiting Earth in a space shuttle / on the moon
  - Read section 6-5 for more on these.

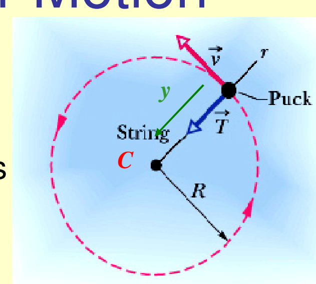
Uniform Circular Motion



- How do you go about solving problems involving Uniform Circular Motion of an object of mass m on a circular orbit of radius r with speed v?
- Draw the diagram for the object.
- Choose coordinate system so that one of the axes points toward the orbit centre C (the y-axis in this case).
- Determine $F_{net,y}$
- Set $F_{net,y} = m\frac{v^2}{r}$

17

Uniform Circular Motion



- A hockey puck moves around a circle at constant speed v on a horizontal ice surface. The puck is tied to a string looped around a peg at point C.
- Force diagram for puck.
- Choose axes
- Determine net force along the y-axis

$$F_{net,y} = T = m\frac{v^2}{r}$$

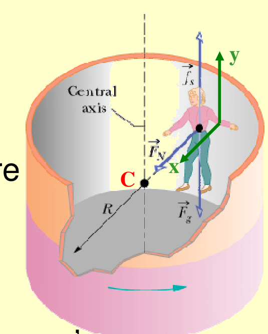
18

Problem

- An astronaut is in the International Space Station, in a circular orbit around Earth, at an altitude h = 520 km and with a constant speed v = 7.6 km/s. His mass is 79 kg.
  - What is his acceleration?
  - What force does Earth exert on him?

19

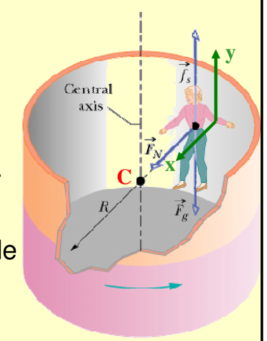
Probleem



- At the fairground, the Rotor is a ride where the participants stand against the wall of the cylinder. It starts to rotate and at a certain speed, the floor falls away. The participant does not fall. Suppose the coefficient of static friction between the person's clothes and the canvas μ_s = 0.40 and the radius of the cylinder is R = 2.1 m.
  - What minimum speed must the cylinder and rider have if the rider is not to fall when the floor drops?
  - If the riders mass is 49 kg, what is the magnitude of the centripetal force on her?
- Draw the force diagram for the object.

20

Probleem

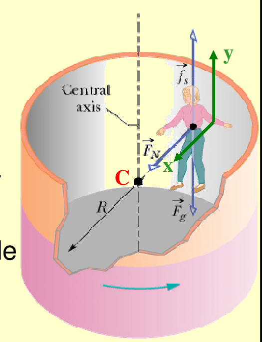


- What minimum speed must the cylinder and rider have if the rider is not to fall when the floor drops?
- If the riders mass is 49 kg, what is the magnitude of the centripetal force on her?

- Choose one of the coordinate axes to point toward the orbit centre *C* (the x-axis in this case).
- Determine net force on rider.
  -
  -
  -
  -

21

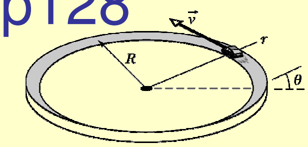
Probleem



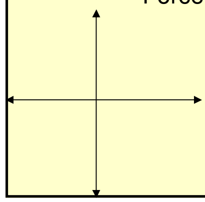
- What minimum speed must the cylinder and rider have if the rider is not to fall when the floor drops?
- If the riders mass is 49 kg, what is the magnitude of the centripetal force on her?

22

Sample Problem p128



- Curved portions of a highway are usually also banked (tilted) to prevent cars from sliding off the highway. When a highway is dry, the frictional force between tires and the road may be enough to prevent sliding, but when the road is wet, the frictional force may be negligible. The figure represents a car of mass m as it moves at constant speed v = 20 m/s around a banked circular track of radius R = 190 m. If the frictional force from the road is negligible, what bank angle θ prevents sliding?
  - v = 20 m/s, R = 190 m
  - Force Diagram



Uniform Circular Motion

- Examples:
  - Rounding a curve in a car – passenger on the centre of the back seat. Car travelling straight. Then driver suddenly turns left around a corner. Passenger – slides to ....?? Car continues around corner – Newton's 2nd law. Centripetal forces is caused by friction on the tires from road. For the passenger…?
  - Orbiting Earth in a space shuttle – you float around the cabin. Why different to car example? You and shuttle in circular motion. Centripetal motion is caused by gravitational force of the Earth.
  - Both uniform circular motion yet sensations differ. Why?
  - In 1 you feel a push of car on your body. The other is a pull on you and the shuttle by gravitational force (much less than on Earth).